

Development and inner environment analysis of advance and eco-friendly cementitious silo for postharvest grain protection and shelf life extension for medium and small-scale farmers

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ABSTRACT

This research addresses small-scale farmers' challenges in developing countries, focusing on post-harvest losses due to inadequate storage systems. Emphasizing food security amid a projected global population of 9.2 billion by 2050, particularly in developing nations, the study introduces an eco-friendly cementitious silo designed for small-scale farmers. The silo incorporates innovative features, including rubber stoppers, herbal lining, pad-locked lids for efficiency and security, and reinforcement with wire mesh and natural fibers for mobility. Temperature and pressure measurements within the vertical silo during laminar filling are explored, considering varying moisture content for grains like wheat, rice, and beans. Results show an upward temperature trend, with beans reaching 49 °C, wheat 42.5 °C, and rice 30.5 °C. The enhanced cementitious silo exhibits a 90% reduction in insect infestation, significant temperature stability, and eliminates grain caking. With flexibility in construction (50–2000 kg), these silos are ideal for storing wheat, rice, beans, maize, millets, and various cereals, offering a sustainable solution for small-scale farmers in developing countries.

1. Introduction

In ancient times, people consumed animals for food but sought alternatives during adverse weather. Around 10,000 years ago, settled agriculture emerged, introducing root vegetables as an alternative to nomadic lifestyles. Grain cultivation and storage practices developed approximately 6000 and 4500 years ago (Levinson and Levinson, 1985). The shift from nomadic to settled living increased dependence on cultivation, leading to the necessity for long-term grain storage. The earliest grain silo construction was discovered in Jordan's valley at Tel Tsaf (5200–4600 BCE) (Rosenberg, et al., 2017). Archaeological findings in India also highlight diverse grain storage methods, including clay bins, chambers, pit silos, storage jars, storage platforms, and granaries (Nag and Kumar, 2023a). These historical practices underscore the importance of careful storage. While modern warehousing systems persist, there is a pressing need to introduce advanced techniques to control post-harvest losses effectively.

Small and medium-scale farmers in developing nations, including India, currently grapple with challenges such as insufficient quality seedlings, fertilizer, storage facilities, and issues related to animals and pests. Agriculture is anticipated to contribute up to 50% of India's GDP, supporting over 80% of the world's population in developing countries,

relying on 500 million small farms that sustain around two billion people (Ruane and Sonnino, 2011). According to a study by the National Bank for Agriculture and Rural Development (NABARD), agriculture supports 10.07 crore families in India, constituting 48% of all households in the nation, with an average household size of 4.9 members in 2016–17. Conversely, the Central Institute of Post-Harvest Engineering and Technology (CIPHET) notes that 4.65%–5.99% of India's primary food crop yield is lost post-harvest due to inadequate grain storage practices, pests, and the absence of storage management technologies, with 52–56% of these losses attributed to poor storage systems (Nuthalapati et al., 2022). This situation forces many small farmers into a cycle of poverty, compelling them to sell their harvest at low prices only to repurchase it at higher prices later due to these challenges. Even attempts at storage are hindered by grain weevils, causing significant damage to the grains (Puthur et al., 2019) and exacerbating the challenges farmers face.

Addressing these issues is crucial for global food production, necessitating significant investments in the agricultural industry. Countering challenges like inadequate storage facilities, postharvest losses, and distribution inefficiencies requires a comprehensive approach. Improving storage methods, implementing value-added marketing, and enhancing distribution strategies are essential to boost agricultural

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output. For those designing storage bins, stresses on silo walls and the base, influenced by internal and external factors, are critical parameters. Factors determining storage conditions, especially grain temperature and moisture content, play a pivotal role in cereal storage, closely impacting grain quality (Uthayakumaran and Wrigley, 2017). Cereals stored in bins are never uniformly moist due to their capillary-porous structure and are highly susceptible to water absorption and desorption. The diffusion of water within a silo results in seed swelling, increasing loads on the bin's walls and base. Additionally, biological processes like self-heating seeds contribute to thermal loads that designers and technologists must consider (Lacey et al., 1991).

Ambient temperature influences cylindrical bins, potentially causing wall deformations. Selecting materials for silo construction must consider application, location, and temperature. Cold temperatures impose heavy loads on grain-filled bin walls, necessitating careful design considerations. In experiments by Chandra and Berntsson (2002) a 0.9 m diameter, 1.2 m height bin made of 0.8 mm thick sheet aluminium filled with grain showed mean load increases of 0.22 kPa/°C, 0.36 kPa/°C and 0.38 kPa/°C across three temperature cycles in the range of +10 °C. A mathematical model based on the finite element model (Do et al., 2019) demonstrated that theoretically calculated load values exceeded measured values during temperature changes. Montross and Maier (1998) emphasized the precarious nature of presumed safe moisture content for prolonged grain storage due to water migration or redistribution within bins. Their model revealed water migration in silos of all sizes, especially noticeable in smaller bins. Kebeli et al. (2000) highlighted the significant impact of temperature changes on load directions and the horizontal-to-vertical load ratio in bins filled with wheat grain at 10% moisture content. Using a two-dimensional finite element theory, their analysis explained the distribution of loads induced by thermal processes. They also demonstrated the occurrence of seed swelling and extremely high loads in bin construction during grain moisturizing in the silo, devising a mathematical model based on a proportional relationship between the volume increase of a single grain and the absorbed water amount.

Zhang et al. (1998) stressed the importance of ASAE's EP 433 standard, advocating for a moisture content range of 1–2% during grain storage in silos to prevent high loads from seed swelling. Subrot Panigrahi et al. (2020) investigated various parameters affecting grain temperature during loose storage, including bin diameter, layer height, silo wall material, silo shape, and grain mixing with a platform auger. Larger silo diameters caused a time delay in temperature changes near the bin centre, while narrow and tall bins maintained lower temperatures than short bins with large diameters. Material composition influenced temperature, with zinc-coated sheet silos exhibiting higher temperatures than white-painted ones. Although bin shape and grain mixing had minimal effects on temperature during longer storage times, grain mixing effectively reduced grain temperature near the silo centre by about 10 °C immediately after the process. Jian et al. (2005) developed a three-dimensional model for temperature distribution within a grain storage bin, requiring initial temperature distribution measurements and meteorological data. The model revealed temperature variations of 5–15 °C between sunlit and shaded sides, providing insights not achievable in two-dimensional models. The application of the model relies on knowledge of thermal properties for grain, bin wall material, soil, and air. The author established high consistency between simulation and experimental results using real silos filled with rape seeds and barley.

Water condensation on bin walls and roofs poses a common issue in storage challenges. Salleh (1990) observed condensation during white rice storage, conducting studies where rice was stored at 10 °C and 13.5% moisture for six months in a cylindrical bin at 25 °C outer temperature. Bartkowiak et al. (2019) noted water condensation during barley storage, attributing it to barley respiration, causing water desorption and increased mean moisture content, leading to elevated wall loads and grain temperature due to self-heating. Horabik and

Molenda (2017) explored the impact of moisture gradients on load distribution on bin walls, highlighting wheat grain's highest load. In contrast, traditional grain silos (Kothi) made of clay or mud reinforced with agricultural waste fibers serve as prevalent storage solutions in Indian villages (Salamon, 1995). These silos, often cylindrical or rectangular, symbolize prosperity and security for farmers' families. However, technological advancements are crucial to enhance even modest cementitious silos for effective crop storage. This study establishes hermetic sealing for cementitious silos, making them suitable for fumigation and eliminating rodents and insects. This advancement allows padlocking grain storage areas to prevent theft and unauthorized access. Considerations for the technology include simple relocation, low cost, quick sanitation, and risk-free silo emptying.

The grain storage is a part of our Indian grain marketing supply chain for maintaining and coping with the grain supply from the high production year to the low production year or one harvest season to another dormant season. Due to the variation of rainwater in India, the agriculture yield is not the same. Another major purpose for storing food grains is to reduce losses from rats, insects, rodents, pathogens, and other pests. However, 52–56% of the total post-harvest loss contribution is still due to poor storage facilities in India (Nathulapati et al., 2022). Even if there are storage facilities, chemicals are used to preserve the grains, harming our health and environment. The grains are preserved due to chemicals, but we are also getting many types of diseases that are not immediately visible due to the use of those grains. Over the past 50 years, chemical pesticides have been widely used to lessen the amount of grain damaged by insects. However, this practice has resulted in several environmental and human health issues (Nag et al., 2021). These issues drove scientists to seek solutions that would produce less negative side effects, and they are trying their effort in several ways. One of which plant/Herbal based pesticide had been used in the different storage system (Bezabih et al., 2022). Before applying these eco-friendly technologies, cereal's temperature, moisture, and oxygen content should be controlled before storing it because these are the initial reasons for the spoilage of the food grains.

Insects and rodents are infesting and destroying our commodities. Various insects and rodents pose threats to crops at different stages of development. For instance, the Maize weevil (*Sitophilus zeamais*) and Rice weevil (*Sitophilus oryzae*) tend to infest commodities like maize, sorghum, and rice. Their infestation is low during the pre-harvest period but increases significantly in the shelled stage. *Acanthoscelides obtectus* moderately infests beans at a low rate before harvest and intensifies during the shelled stage. The Large Grain borer (*Prostephanus truncatus*) primarily targets maize, with minimal infestation during the milled kernel stage and higher levels during the unshelled stage of crop storage. Additionally, *Rhyzopertha dominica*, *Tribolium castaneum*, along with other secondary beetles, *Lasioderma serricorne*, *Sitotroga cerealella* (Angoumois Grain moth), and rodents such as rats and mice, cause damage or infestation across various cereal commodities at different stages of pre-harvest, unshelled, shelled, and milled kernels (Banga et al., 2018). Government of India Ministry of Agriculture & Farmers Welfare Department of Agriculture & Farmers Welfare Directorate of Economics & Statistics suggest the grain production loss due to poor storage (MoA&FW).

Therefore, the manuscript explores the historical evolution of grain storage from ancient times to modern challenges faced by small-scale farmers. It emphasizes the need for advanced techniques to control post-harvest losses, especially in developing nations like India. The Central Institute of Post-Harvest Engineering and Technology highlights significant food crop wastage due to poor storage practices. The paper delves into the complexities of designing storage bins, considering factors such as temperature, moisture content, and grain quality. It sheds light on experiments demonstrating the impact of ambient temperature on cylindrical bins, emphasizing the importance of material selection for construction. Mathematical models and standards are presented to address challenges like water migration and seed swelling. The study

also covers the unique challenges of traditional grain silos and proposes technological advancements for effective crop protection. The exploration of water condensation issues and the development of hermetic sealing for cementitious silos are discussed, underlining the constant need for innovation in agricultural practices.

2. Materials and methods

This research was released after a thorough literature review and investigation of the issue in a rural area to focus attention on the difficulties encountered by small-scale farmers while attempting to store grains in an eco-friendly manner. The author focused specifically on the problems involving the traditional storage systems of grain cementitious silos. As a result, the cementitious silo technique was created to safeguard grains and reduce postharvest losses efficiently. For small and medium-sized farms, this study provides a novel dynamic model design of a movable cementitious silo with full air-tight and padlocking capabilities. Together, these two goals hunger and poverty of the developing countries and the world will greatly be benefitted from this effort.

The construction of the advanced cementitious silo required careful consideration of the following factors throughout the design phase. (i) Fabrication of rigid structure to bear the design load (ii) Structure should be cheaper and easily maintainable (iii) Easy for shifting the structure (iv) Fireproof structure (v) Provide sealing pads for airtight padlocks (vi) Reducing extreme temperature fluctuations can boost storage quality (vii) Applied herbal pest controlled practices (herbal pest liner applied on the inner wall of the silo and bio-fumigation) (viii) Lightweight structure that makes loading and unloading simple. The factors that determine the construction materials and design may be stated as the considerations considered before designing of the advanced cementitious silo.

2.1. Fabrication of advance cementitious silo

Traditional grain silo (Kothi) is made up of mud with post-harvesting wastage like a wheat husk. This husk is used for reinforcing the mud and enhancing the strength of the silo. However, this method is not very reliable and needs major improvement. In this advanced era, innovation is at its peak. So, for advanced fabrication of traditional grain silo, lime powder, fly ash, natural fibers (Corn husk, coir), wire mesh, basil powder, neem leaf powder, and cow dung were used. Skilled hands manufactured this traditionally advanced eco-friendly grain silo (Kothi). The grains inside this advanced silo were protected and safe from insects and pathogens. Since their fabrication involved pure scientific and eco-friendly concepts that were not used till now. The fly ash and limestone

powder were well mixed with water and reinforced with natural fibers and wire mesh for sturdy and light weight design. A manual hand layup process prepares it. Indian farmers generally prepared in their houses. Fly ash and lime powder make the wall dry prevent from moisture and provide an additional safeguard of the stored grains. The inner wall of the silo was additionally plastered with a paste of herbal powders (basil, neem leaf, neem seeds) and cow dung in their adequate proportion. Although this additional herbal plastering on the inner wall of the silo makes the pesticidal environment of the advanced cementitious silo. This plastered inner wall prevents insects and pathogens in the grains. Traditional mud silo is made as advanced silo by modifying its construction and introducing advance features. Reinforcement with natural fibers and wire mesh make it lighter and strong, whole structure made movable, introduce inlet and outlet air tight pad locking system for hermetic environment inside the silo, introducing modified discharge unit for easy outlet system etc. Fig. 1 depicts the pictorial view of the developed advanced cementitious silo. The miscellaneous description has been further elaborated.

2.2. Structural and technological advancement of cementitious silo

For small and medium-sized farms, the cementitious silo technique offers a practical solution to the problem of grain spoilage after harvest. This method makes long- and short-term grain storage possible by preventing pathogen damage, animal, and insect pest. Traditional cementitious silos are often manufactured in the shapes of cylindrical, square, or rectangular prisms of variable diameters out of clay soil or termite mounds of varying thicknesses. In contrast, advanced silos are constructed from lime and fly ash in a 2:1 ratio, with the addition of reinforcement from natural fibers and wire mesh to ensure a strong and lightweight construction. The wall thickness of 50 mm was maintained all around the cylindrical structure. Clay and fly ash are inexpensive, geopolymer for synthesis of green wall. Having high compressive strength for load-bearing capability (Nag and Kumar, 2023b). Moreover, fabricated cementitious silo inner wall plastered with herbal pastes of basil, neem, and cow dung in a significant proportion.

2.3. Silo shell design and conception

The silo wall was designed to be cylindrical so that it could withstand the pressure induced by grain loading by developing circumferential and hoop stresses. The drum's strength may be maximized at a minimal thickness using a cylindrical shape since it takes up minimum space and requires fewer joints during assembly. Additionally, they facilitate silo loading and unloading with less effort. Based on equations (1) and (2),

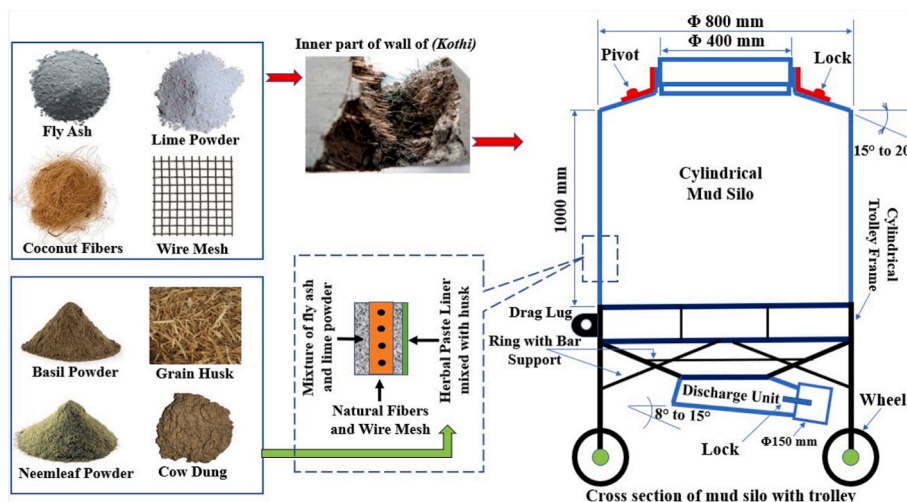


Fig. 1. Fabrication of traditionally advanced cementitious grain silo (Kothi).

the volume or quantity of grain stored was used to calculate the silo's wall height and diameter.

$$V = \pi r^2 H_1 + \frac{1}{3} \pi r^2 H_2 \quad (1)$$

$$\text{But, } V = \frac{m}{\rho} \quad (2)$$

Where V = Volume of the silo (m^3), m = mass of the stored product (kg), H_1 = Height of the silo wall (m), r = Radius of the silo (m), ρ = Density of the stored material (kgm^{-3}), H_2 = height of the hopper (m)

Bwambale et al. (2020) investigated the distribution of pressures caused by charging, discharging, and resting states of the silo. Janssen's equation was used to get the vertical pressure in equations (3)–(5).

$$P_v = \frac{\gamma R}{\mu K} \left(1 - e^{\frac{\mu K x}{R}} \right) \quad (3)$$

$$R = \frac{D}{4} \quad (4)$$

$$K = \frac{1 - \sin \phi}{1 + \sin \phi} \quad (5)$$

Where, P_{Ver} = Vertical Pressure (Nmm^{-2}), R = Hydraulic radius of horizontal cross-section of storage space (m), μ = Coefficient of wall friction, x = Depth from the surface level of stored material (m), K = Lateral to vertical pressure coefficient, γ = Weight per unit volume for stored material (Nmm^{-3}), D = Diameter of the silo (m), ϕ = Angle of internal friction ($^\circ$).

Equation (6) calculated the horizontal pressure, P_{Hor} (Nmm^{-2}), of a unit area at a distance x (mm) from the free surface.

$$P_{\text{Hor}} = K P_{\text{Ver}} \quad (6)$$

With the help of Equation (7), we determined the friction force, P_f (Nmm^{-2}), of a unit area of the silo wall at a distance x (mm) from the assumed free surface.

$$P_f = \mu P_{\text{Hor}} \quad (7)$$

Equation (8) was used to calculate the static unit pressure, P_{Static} (Nmm^{-2}), normal to the surface at a depth of x (mm) below the surface of the stored material.

$$P_{\text{static}} = P_{\text{Hor}} \sin^2 \alpha + P_{\text{Ver}} \cos^2 \alpha \quad (8)$$

Equation (9) (Carson et al., 1993) was used to determine the hoop stress, σ_{Hor} , that would result from a concentric discharge in the silo.

$$\sigma_{\text{Hor}} = \frac{P_{\text{Hor}} D}{2t} \quad (9)$$

Carson and Jenkyn (1993) provided equation (10) for calculating the silo wall thickness based on buckling stress and the hoop stress at the cylinder wall.

We may express the critical elastic buckling stress, σ_{Critical} , as

$$\sigma_{\text{Critical}} = \frac{E}{\sqrt{3(1-\nu^2)}} \times \frac{t}{r} \quad (10)$$

Where σ_{Critical} = Critical elastic buckling stress (Nmm^{-2}), E = Elastic modulus (Nmm^{-2}), t = Cylinder thickness (mm), r = Cylinder radius (mm), ν = Poisson's ratio (unitless).

2.4. Hopper design and conception

The design of the hopper was made conical because that form allows for the most efficient flow when subjected to gravity. The angle of repose of the stored grains should be at least 5° more than the angle of repose to facilitate smooth flow, as stated by Beakawi and Baghabra (2018). The

angle of repose for grains lies between 15 and 30° , as determined by field measurements (Bhadra et al., 2017). Equations (11) and (12) (Nortje, 2002) were used to get the meridional tensile and circumferential hoop stress, respectively.

$$F_m = \frac{P_{\text{Ver}} D}{4 \sin \alpha} + \frac{W}{\pi D \sin \alpha} \quad (11)$$

$$F_t = \frac{P_{\text{Hor}} \sin^2 \alpha + P_{\text{Hor}} \cos^2 \alpha}{2 \sin \alpha} \quad (12)$$

Where, W = Weight of the stored grains in the hopper, F_t = Tangential force due to circumferential hoop stress (N), F_m = Meridional tensile force (N), D = Diameter of the top of the hopper (m).

2.5. Design of the support frame

The trolley support type framework was built to withstand the stress of the stored material. Compressive stress in the bearing frame may be determined using equation (13) from Khurmi and Gupta (2005).

$$\sigma = \frac{\text{Applied Load}}{\text{Cross sectional Area}} \quad (13)$$

2.6. Moisture and layer arrangement effects on silo conditions

Experimental assessments were conducted to observe temperature and pressure variations during wheat, rice, and beans storage. Varied assumptions regarding grain moisture content were applied to individual silos. The model chamber featured two-grain layers with differing moisture levels. In the initial phase, the upper layer had a moisture content (W_u) of 18% w. b., while the lower layer (W_l) was 9% w. b. This configuration was reversed in the second phase. The study aimed to assess how the moisture content and arrangement of layers impact the temperature and pressure distribution of the fabricated cementitious silo. The investigation comprised 3rd day to 25th day of cycles. The investigation utilized the laboratory setup depicted in Fig. 2(a), featuring a cylindrical silo with a cylindrical height of 1000 mm and a diameter of 800 mm. The stand incorporated equipment enabling temperature measurements at 40 different points placed at the level of 100, 200, 300, 400, 500, 600, 700, 800, 900 mm at different distances from the silo axis i.e., 0, 100, 200, 300, and 400 mm. Thermocouple and digital temperature gauges recorded temperature values at specific points, while strain gauges connected to a load indicator measured grain pressure against the silo wall. The upper section of the silo was tightly enclosed using a sealed and insulated rubber lock pad cover, and the temperature within the silo cover was consistently regulated at 16°C through a water supply controlled by the thermostat. Refer to Fig. 2(b) for the arrangement of measurement points.

The silo was loaded with grains having a specified moisture content (W_l) up to a height of 500 mm. After precise surface levelling, it was filled with grains with W_u moisture content up to a height of 1000 mm. The grains were moistened with water and hermetically sealed in plastic barrels for three days to achieve the desired moisture content. Stirring of the grain material was conducted every 8 h to ensure a uniform distribution of moisture. The grain barrels were stored in a controlled climate chamber at 16°C . Measurements of grain pressure against the silo wall and grain temperature at specific points within the silo were taken once daily.

2.7. Design and implementation of grain silo

2.7.1. Portability

In order to ensure that people in poor nations always have access to nutritious food, it is essential to support farming on a smaller scale. The problem of grain postharvest losses can be partly addressed by making transportable storage available (Mhlanga et al., 2010). A circular metal

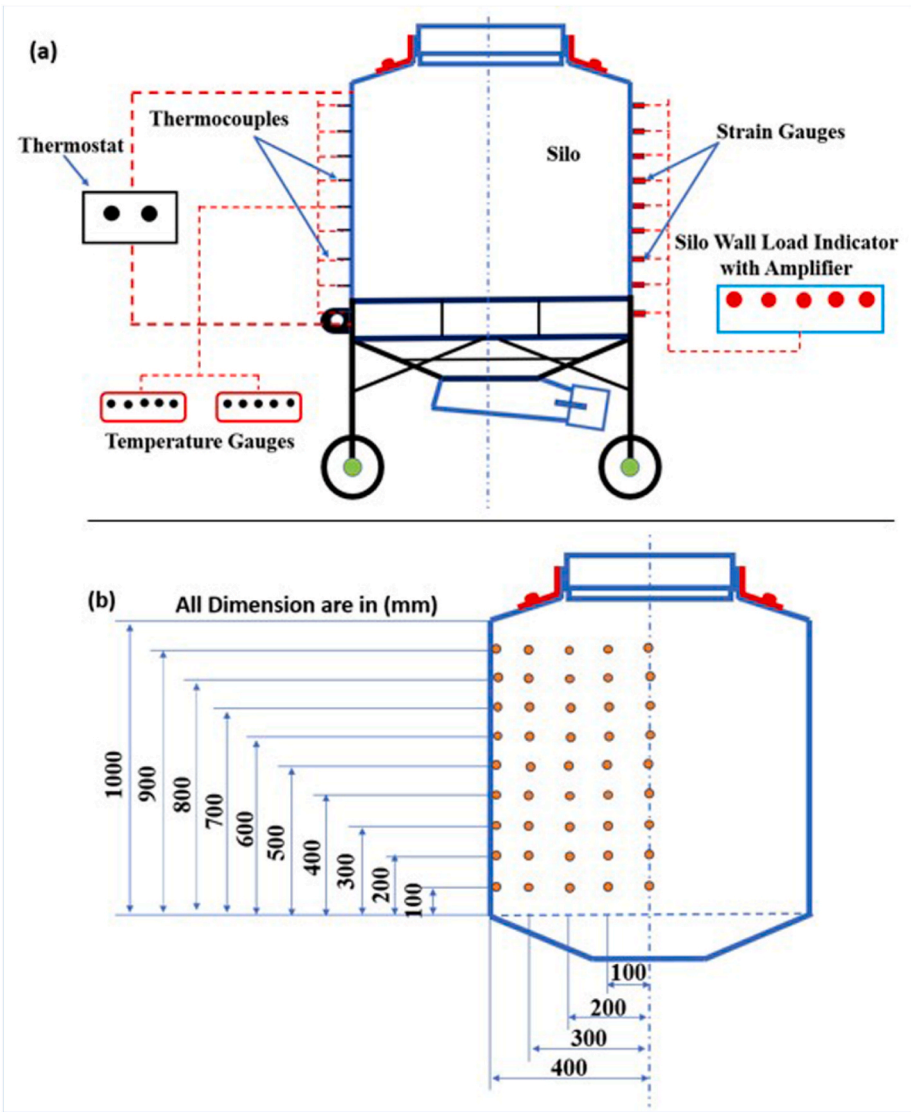


Fig. 2. (a) Illustration of silo testing station (b) Configuration of measuring points in the silo.

frame-stand on four wheels has been constructed to support the cementitious silo, which also has metal-angles, columns, bars, and ring-beams for structural stability. The old, heavy construction and permanent concrete platforms, wooden or metal pallets supports for the flat bottom silos were replaced by this advanced cementitious silo. Moreover, the wheels have been constructed for moving the silo simply if the need arises, since destroying and rebuilding concrete platforms to modify their position is not an economical approach given that the manufactured advance silo may endure for more than 20 years with proper maintenance. Pallet-based silos are very difficult to relocate, regardless of whether they are empty or full of grain.

Fig. 1 depicts a graphical representation of a manufactured cylindrical mobile cementitious silo for small-scale farming. The stated dimensions can hold 500–600 kg of grains such as rice, maize, wheat, guinea corn, millet, and beans. The size of an advanced silo can range from 50 to 3000 kg, depending on the location and the demands of the population (Manandhar et al., 2018). A drag lug has been welded on the mobile cementitious silo so that it may be dragged behind a vehicle or towed at a moderate pace. For the lesser sizes of these silos, wheels may not be necessary, and they may be supported only by a metal frame. If you want to prevent accidents involving the silo’s collapse, build your mobile cementitious silo broader than it is tall, as shown in Fig. 1. On the other hand, the silo’s wheels structure and base shouldn’t be too high

above the ground. Depending on the silo’s dimensions, the wheels will be as broad as feasible to prevent the vehicle from being mired in the soil. Wheels are optional if the unit can function properly on a metal stand frame alone.

2.7.2. Dimension of the silo components

Using equations (7) and (8), it was calculated that the diameter of the silo wall was 800 mm, and the combined heights were 1000 mm and 300 mm. The height of the silo is ideal for loading and unloading grains, and the diameter is just enough for storing them inside. As determined by equations (9) and (12)–(14), Table 1 records the static design, frictional force, vertical, horizontal, and pressure applied to the silo wall as depicted in Fig. 3. The silo wall thickness was estimated to be 40.45 mm

Table 1
The vertical, horizontal, static design pressures and the frictional force in the silo.

Depth (x) from the Top (mm)	P _{Ver} (Nmm ⁻²)	P _{Static} (Nmm ⁻²)	P _{Hor} (Nmm ⁻²)	P _f
1000	0.004593	0.002965	0.001431	0.000691
1150	0.004890	0.003308	0.001658	0.000795
1250	0.005187	0.003651	0.001885	0.000899

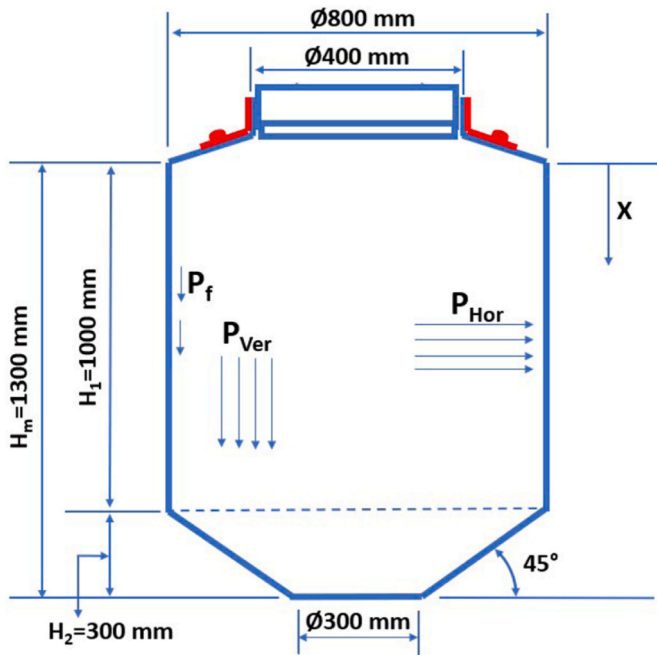


Fig. 3. Pressure distribution on the cementitious silo wall.

by equations (15) and (16) in order to sustain all the predicted pressures. However, during the silo's construction, a conventional thickness of 50 mm was chosen. To provide smooth solid flow during silo discharge, a hopper angle 45° greater than the grain angle of repose was selected. Equation 20 provided the 300 mm minimum exit diameter for the hopper. Equation 19 was used to estimate the support frame design to estimate the total compressive stress of 3.5321 Nmm^{-2} from the weight above the support. The outcome is much lower than the wire mesh and natural fiber-reinforced cementitious composite's compressive stress, which justifies the material's application. For simple unloading of the grain from the silo, four $40 \times 40 \times 3 \text{ mm}$ angle bars were chosen, each measuring 0.9 m height based on the typical height of a person's waist.

2.7.3. Hoppers design of the mobile cementitious silo

The cylindrical movable cementitious silo has two concentric hoppers, one at the top and one at the bottom; these hoppers are truncated cones and are very important. There is a potential for downward deflection of a silo with a flat top during grain loadings, and any rainfall that happens to fall on top might potentially collect and may affect the storing condition. When emptying a silo, a hopper at the bottom that has a concentric discharge is essential since the grain is expelled under its own weight (Saleh et al., 2018). Flat-bottom traditional mud silos can be dangerous since tilting or raising up one end can release all of the grain, but this can also cause the silo to break, twist, deflect, or even bow inward at the base. Due to bulky structure of the traditional mud silo people generally avoid tilting the mud silo and left some grains inside the silo, this is one of the draw back of the traditional mud silo as shown in Fig. 4. This major problem has also been addressed in the advanced cementitious silo. The empty bin is essential for various purposes like maintenance, changing the storage material, fumigation, etc.

For this reason, the bottom hopper features a custom discharge unit. The discharge unit tilted at 15° – 30° for the horizontal.

Equations (14) and (15) were used to establish the hopper's minimum diameter (Chase N.D).

$$d = H(\theta) \frac{CAS}{\rho g / g_c} \pi r^2 \quad (14)$$

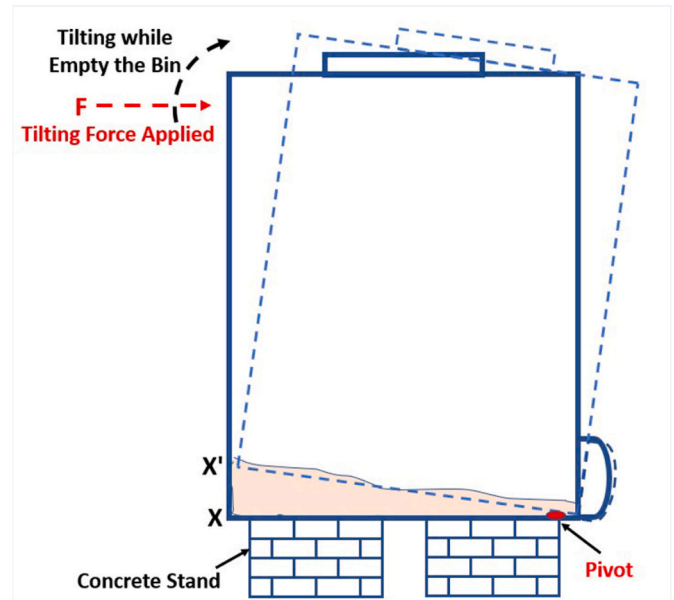


Fig. 4. Problem during emptying the conventional silo needs tilting.

$$H(\theta) = 2 + \frac{\theta}{60} \quad (15)$$

Where.

CAS = Critical applied stress which is a function of the material's flow function and

Flow factor θ = Semi included angle.

CAS was calculated to be 2.8766 using the design chart for symmetrical slot outlet hoppers for solid flow (Nakashima, 2010; Woodcock et al., 1987).

Further, a discharge unit's outlet was introduced at the intersection of the tip of a truncated cone. The inlet and outlet of the silo's should be concentric with structure centre. As shown in Fig. 1, the discharge point is not supposed to be pointing straight down but at an angle of 8° – 15° off the horizontal plane. The discharge point is often tilted to minimize grain pressure and maintain complete control over the discharge process (Vidal et al., 2005). The grain entrance and output points should be symmetrical about the silo's geometric centre if at all possible.

2.7.4. Inlet cover with rubber sealing

The compact cementitious silos are ideal for storing grains out of the reach of rodents and insects, and it is also a best practice for storing food grains for personal use. However, the traditional cementitious silo structures were made up with soil and this was used for a short duration of storage (Mobolade et al., 2019). Grain losses can be substantial if silos are not sealed properly. Hence in advance cementitious silo a rubber sealing stopper provided under the inlet cover of the silo. Whereas, in traditional cementitious silo there was no any such provision for repeated use of stored grain. However, in advance grain silos, a rubber sealing stopper provided to blocks every possible gap between the body of the silo and the lids. Because ambient air may enter in to the silo through minor gaps, and may spoil the stored grains. The Advanced cementitious silo can be open and close very easily during pouring the grains or delivery the grains through inlet and outlet respectively. To provide optimal grain security from rodent and insect pest assaults, the silo must be hermetically sealed. The most cost-effective option was to install rubber stoppers under the grain inflow and discharge lids of the silo, which worked flawlessly during fumigation and long-term grain storage. Fig. 5 depicts a cross-section view of a rubber stopper squeezed

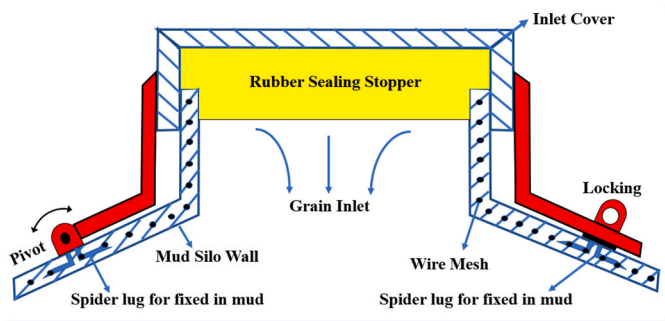


Fig. 5. Sectional view of inlet cover sealing unit.

in between the silo lid and the silo top edge of the grain Inlet aperture to achieve the hermetic sealing. Hermetic grain storage disrupts insect, fungal, and grain respiration by eliminating oxygen and increasing carbon dioxide in the storage environment. When the oxygen level in the storage system drops to 3% or below, all the insects will perish. In addition, when oxygen levels drop below 1%, fungal growth stops (Quezada et al., 2006). It is only viable when this particular airtight silo is closed, and the grain is in an enclosed and regulated environment.

2.7.5. Pivoting and padlocking the silo

This cementitious silo improvement features a fastening capability with a rubber padlock at both the grain inlet and the outflow. Fig. 6 shows the cross-section top view of the bottom outflow discharge unit. This strengthens the safety measures and makes it impossible for children to access or destroy the grain. The silo cover (Fig. 5). is designed to rotate on a pivot or hinge, and the locking point is located on the opposite side of the pivot. The spider lug design for both the end of the pivot and locking unit makes them fixed in the cementitious wall. However, it is also important to have rubber stoppers beneath the cover and padlocking capabilities, even though the coverings may be built in various methods and shapes.

2.7.6. Cost of the advanced cementitious silo

A moveable cementitious silo costs a little more to construct than a typical flat-bottom cementitious silo. Mobile cementitious silos have had a rubber-stopper installed under the grain coverings, a padlocking point installed, the flat top and bottom converted to hoppers, the structure reinforced with wire mesh, pivot frame, locking unit, and herbal plastering also applied inside the wall. This causes a slight but noticeable rise in price, which is feasible. The only difference is the addition of wheels and a metal frame to the stand. This drives up the price because it eliminates the need for concrete footings, wooden or metal pallets and makes the entire construction portable. As silo capacity grows, the price per kilo of grain stored decreases. Table 2 displays the price at which advanced cementitious silos had been fabricated. All the critical items have been purchased through the India Mart website. The actual price cost accounted for this silo’s construction may be decreased if the

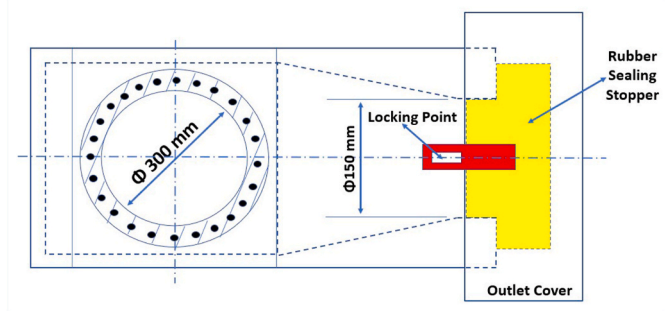


Fig. 6. Section of top view of bottom discharge unit/outlet.

Table 2

Cost estimation of fabricated advanced cementitious silo.

Material/Process	Price	Qty.	Cost
Fly ash	2000 ₹/Tonne	200 kg	400 ₹
Lime powder	3500 ₹/Tonne	400 kg	1400 ₹
Coir fiber	15 ₹/kg	60 kg	900 ₹
Wire mesh (φ of wire is 2 mm)	52 ₹/kg	20 kg	1040 ₹
Herbal Powder (Neem and Basil)	75 ₹/kg and 100 ₹/kg	3 kg + 3 kg	525 ₹
Pivot Unit	300 ₹/unit	2 unit	600 ₹
Locking Unit	250 ₹/unit	2 unit	500 ₹
Trolley wheel (Heavy Duty)	500 ₹/unit	4 unit	2000 ₹
Rubber Pad	150 ₹/kg	4 kg	600 ₹
Steel Channel	75 ₹/kg	100 kg	7500 ₹
Welding	800 ₹	—	800 ₹
Labor Cost	700 ₹/day	2 day	1400 ₹
Total Cost = 17,665 ₹			

production unit may increase. Only the moveable trolley contributed 58% of the overall cost of the advanced cementitious silo construction. Equipment depreciation, utilities, and shipping costs are not included. Depending on the specific of each nation or location, but overall prices may be reasonable for this advance cementitious silo.

2.7.7. Silo performance analysis

To ensure proper sealing of the silo, a candle test was performed. A candle was placed in the silo and watched (Kimani et al., 2018). A stopwatch was used in three separate trials to measure how long it took from when the silo was shut to the time candle burned out.

Dried grain samples with a wet basis moisture content of 13.9% were placed into the silo and inspected for insect infestation. A small digital hygrometer was used to determine the relative humidity because of its convenience, precision, and reliability (Tubbs et al., 2017). Ten (10) samples of 0.25 kg each were prepared and placed in the silo with a burning candle. The mean number of weevils found alive per 1 kg of grain was calculated. After 35 days, the grains were removed from the silo and inspected for the presence of living insects in each sample. Equation (16) was used to determine the efficiency of the silo and how well silos protect grain from pests. This approach was used because scientists have shown it to be a reliable and easy way to assess the level of insect infestation inside a sealed silo (Kimani et al., 2018; Yusuf and He, 2011).

$$Efficiency = 1 - \frac{\text{no of live in sec ts per kg after loading in the silo}}{\text{no of in sec ts per kg before loading in the silo}} \quad (16)$$

The process was repeated using the same grain batch (same insect infestation). During this experiment, the burning candle was kept separate from the grains, and after 35 days of storage, a count of the number of active insects per kg was taken. The inside silo temperature was monitored using thermocouples and a TC-800D data recorder at 8-h intervals for 15 weeks to assess its resistance to temperature fluctuations. A digital thermometer was used to measure the ambient temperature. The average temperature of the grains before processing was 21.32 °C. A minimum of 30 min of agitation every week was performed.

2.7.8. Storage humidity reduction

The condensation in advance cementitious silo can’t occur if the air temperature drops below its dew point. However, this has no any significant effect on the grain within the silo since the advance silo wall built with natural fibers and lime powder, which function as a good insulator between the grains and the outside environment, especially when there are big temperature variations, such as at night or during the sunny days. Although grain is such a good heat insulator, any temperature changes within the grain mass will be negligible, and the silo walls can’t be cooled quickly. The air within the silo will lose its heat mostly by convection, which will be significantly slower, resulting in a negligible change in the grain’s moisture content. As a result, even if there is a small risk of condensation, the amount of water deposited would be

quite minimal and remains very low, provided the silo is sealed completely (Rosentrater, 2022).

According to Nigeria's seasoned silo users, condensation issues are never an issue with airtight facilities. Condensation is less of a concern in sealed small cementitious silos than in unsealed ones, provided the grain is within the acceptable storage moisture content threshold. Grain with a moisture level of 14% or less and seed with a moisture content of 12% or less are ideal for long-term preservation (Quezada et al., 2006).

2.7.9. Determining the moisture content of grain

Grain drying is the initial process step essential for successful and long-term storage. Losses might reach 100% if drying is not done properly. Sun drying is a common and traditional way to preserve grain before storage. In this method, the moisture of the grain is removed before facing it to sunlight for a long duration before storing it. Approx. 93% of Indian farmers follow this eco-friendly technique. The scientific reason behind it is that the sun decimates insects and pathogens and enhances the latency period of grains. Sunlight reduces the moisture from the grain and can maintain up to 10–12% moisture, which is good for properly storing the grains (Kumar and Kalita, 2017).

This method of drying grains before storage is advisable for farmers who follow it. Drying times vary widely based on factors like grain variety and climate. Safe grain storage requires a moisture content of 14% or less (Fields and Korunic, 2000). India and other developing countries expose their grains for drying before the sun on agricultural by-products, tarpaulin mats, etc. (Mobolade et al., 2019). However, due to soil moisture, spreading grains on bare ground is not recommended, since the grain would be gathered with certain foreign materials like gravels and dust. Most small-scale farmers in developing nations rely on age-old techniques to determine if the grain is dry by feeling it with their hands, breaking some grains, and watching it with their eyes.

Small farmers can benefit from innovative, low-cost techniques of checking the moisture content of their grain before storing it. Grain moisture testers and regular weighing scales are also options for finding the ideal moisture level in the grain. Small farmers should be trained to these practices through aware farmer's groups, local farmers information centers, or any local kisan Salahkar Mitra.

2.7.10. Electronic grains moisture tester

Available grain moisture testers come in various sizes and forms, from digital to analog, and can be powered by a different power outlet like rechargeable lithium-ion batteries, or durable AAA batteries. It is a bit costly and requires a skilled person to check the grain's moisture. General people can also understand how to operate the electronic grains moisture testers once explained. Fig. 7 (a), (b), and (c) shows display and key function, electronic digital grain moisture tester (M/s- YIERYI International instrument and meter factory), and their operating codes for different grains respectively. This is used in a very simple manner by

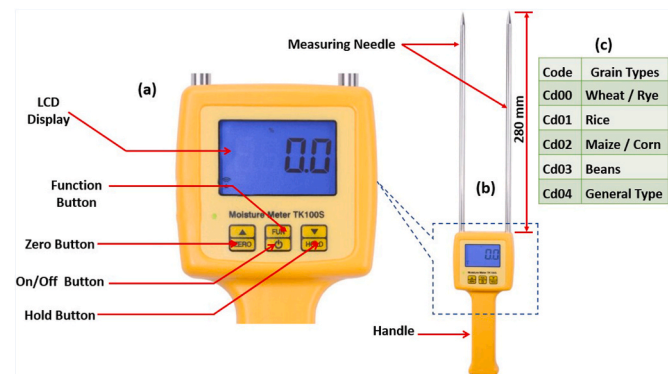


Fig. 7. Grain Moisture tester (a) Display and Key function (b) Grain moisture tester (c) Code used for testing a variety of grains.

only inserting the measuring needle into the stored grain. It measures moisture content ranging from 5 to 35% and operating conditions (temperature 0 °C–60 °C and under the humidity environment range of 5%–70% relative humidity).

Before using the moisture tester, some grain was extracted from the outlet of the advance silo on the frequent interval of when the grain was charged in the silo. Extracted grains was putted into cylindrical or conical bucket, and switched on the moisture tester. After that press the function button according to grains variety (function code provided along with this tester). Putting the needle into the grain filled bucket and measured the average value of the reading at different inserting position of the needle. After inserting 4 to 5 times, the reading was stable. The recorded reading under different condition on frequent days have been listed in Table 3. However, many small farmers are unaware that various grain moisture testers could be useful for ensuring safe and secure storage. Hence it is primary requirement for spread awareness among the small scale farmers for new technology advancement in storing the grains.

2.7.11. Electronic digital weighing machine for grain dryness testing

An electronic digital weighing machine is one of the simplest and most cost-effective ways to determine, measure, and ensure perfect grain dryness. Rural regions, particularly in India and other developing countries like Bangladesh, Pakistan, Bhutan, Nepal etc., use electronic digital weighing machine (as shown in Fig. 8.) rather than electronic grain moisture testers. Since the method of weighing through electronic machines is more familiar, they also use it for various purposes in daily life. Additionally, the findings of digital weighing scales, whether electrical or battery-operated, can include up to three significant digits, such as 2.678 kg. However, a spring-loaded dial scale can be used if there is no digital option available, and it is very cautious while taking exact measurements, as the spring-based device will eventually wear out and give inaccurate results if used too often. Unlike a grain moisture meter, this weighing scale cannot immediately offer the moisture information available in the grain.

Whether the grain is ready for storage or not depends on whether the grain is completely dry or not. For dryness testing, at first the grain was dried until by visual inspection it is not confirm that grains dried. Further, some sample quantity was taken and placed in the bowl of an electronic weighing machine, and the weight was measured. The same process was carried out for some more grain, and the weight was measured. This process was repeated 4–5 times. Again, another day drying of the same sample grain with main grain in sun exposure measured the weight through the same process. The same measuring process was performed. If the grain weight at today's weighing measurement is the same as yesterday's, it has dried sufficiently; otherwise, further drying time is needed. Drying and weighing the same grain samples repeated until the same results are obtained for the two consecutive days test. It indicates that all the water has evaporated, there has been no more weight loss, and the grain is now suitable for storage.

Also, consider the following safety measures: utilize grain samples weighing 2 kg or more to identify any weight drop. For the most accurate results, it is also cautious that not add or remove even a single grain from the samples before testing and drying. Samples should be dried in the same location as the main grain. It is also notable that sample and main grain shouldn't put some in the shade and others in the sun. For consistent dryness, it is best to scatter the samples at a depth equal to the main grain depth. Rotating the grain more frequently and keeping it below 42 °C when sun drying the seed (Yang et al., 2006) is important. When temps rise above it, it is best to dry the grain in the shade. Always weigh and store samples using the same nylon bag or container.

2.7.12. Grain moisture content calculation

The moisture content available in the grain is measured by the weight of the grain, referred to as the Grain's moisture content (MC).

Table 3

Recorded data of fabricated advance silo in grain storage condition (Relative humidity, Ambient temperature, and Average temperature).

Day	Average Relative Humidity (%)	Average Ambient Temperature (°C)	Average Temp (°C)	Moisture Content (%)			
				Rice	Wheat	Maize	Beans
1st	90	31	30	26.2	17.2	15.5	24.7
2nd	90	31	29	29.1	20.6	18.2	27.3
3rd	90	30	27	31.6	22.2	22.1	28.3
4th	90	30	27	18.7	18.5	17.0	12.4
5th	90	30	27	11.3	16.2	12.1	11.2
6th	71	30	25	11.1	13.3	12.5	9.5
8th	68	29	25	10.6	13.0	11.8	9.2
12th	65	25	21	9.1	10.6	11.8	9
16th	55	25	21	9.1	10.6	11.8	9
20th	55	25	21	9.1	10.6	11.8	9
25th	55	25	21	9.1	10.6	11.8	9

**Fig. 8.** Electronic digital weighing machine.

Most commonly, “MC” refers to the “wet basis,” or the overall weight of the grain, including the water (MC_{wb}). Moisture content (MC) sometimes also called dry matter of the grain, and it is commonly used in scientific studies denoted as MC_{db} . The moisture content is expressed in percentage (%). This is the most common approach, and it involves using either an oven or an infrared moisture balance to quantify the amount of moisture present. Secondary method is electronic instrument for measuring the moisture % discussed earlier. Inaccurate assessment of moisture content leads to deterioration and losses in the grain management and marketing process. The following equations were used to calculate the moisture percentage MC (%).

$$MC_{wb} = \frac{W_i - W_f}{W_i} \times 100 \quad (17)$$

$$MC_{db} = \frac{W_i - W_f}{W_i} \times 100 \quad (18)$$

Where, MC_{wb} = Moisture content wet basis in (%), MC_{db} = Moisture content dry basis in (%), W_i = Initial weight in (gm), W_f = Final weight in (gm)

2.7.13. Care and safety measures for the cementitious silo

Before storing grain in the silo, make sure the interior and exterior walls are spotless, check for any holes in the silo’s body, and store only dry grains that meet the necessary moisture level (Bucklin et al., 2019). Grain quality can also be damaged from prolonged exposure to the sun, thus silos need to be placed under a roof or cover to keep out rain and solar heat radiation. There should be no wall contact between the cementitious silo and the wall, especially during the rainy season. The top of the silo should not be used as storage for anything very heavy.

It’s important to properly lock the wheel of the mobile cementitious silo when it’s not in use to prevent it from shifting and causing an

accident. When charging grain into a tall silo, it’s safer to use a chair, table, or single or double-sided ladder than to rest the ladder against the silo wall itself. Also, Wojcik et al., 2010 suggest a cylindrical silo with concentric grain input and grain outflow to provide a balanced grain pressure on the silo walls (Wojcik et al., 2003). Other practices recommended for storing grain include exchanging it with newly harvested grain after one to two years or checking its moisture content frequently to ensure it remains below 9% (Chang and Steele, 1995).

2.7.14. Herbal pesticide and fumigant for storage

Broad-spectrum synthetic pesticides are used on a massive scale, leading to major problems, including pesticide resistance and residue, biosphere contamination, and lifelong health hazards. This has prompted the search for biodegradable pesticides with improved selectivity. However, the insecticide must be effective against the targeted pest while being safe for humans and other mammals, eco-friendly, resistant-pest-free, cheap, and accessible locally. So, the well-known plant neem’s leaves, crushed seeds, powdered fruits, oil, and so on have been utilized as an eco-friendly grain protectant (Talukder et al., 2004). The Neem tree has been used to combat insects in homes, warehouses, and fields. Fumigant made from Neem leaves is both a pesticide and a disinfectant, and it doesn’t harm the environment in the process. In addition to killing pests, this natural fumigant has several deleterious effects on pests, such as preventing them from feeding, laying eggs, mating, growing, etc. Neem fumigant is also useful for preventing pests in grain silos. This organic fumigant is advantageous since it prevents pests from developing a resistance to it and does not leave any residue on the grain (Mondal and Chakraborty, 2016).

The advanced cementitious silo’s airtight storage capacity is obtained by a rubber stopper installed under the silo cover at the inlet and outlet. The airtight storage builds on the assumption that insects would perish due to a lack of oxygen to achieve the intended purpose. Further, the neem paste plastered on the inner wall and neem leaf provided a certain depth of layer of stored grain.

3. Results and discussion

3.1. Performance evaluation of fabricated cementitious silo

The results of the candle test show that the minimum time required before the candle burns out is 22 min, 25 min, and 19 min, respectively. The silo’s oxygen had been burned for, on average, 22 min when the candle went out, and there was little or no oxygen entering the silo, suggesting an airtight environment. The silo’s effectiveness in reducing insect infestation was evaluated after 35 days of grain storage, with the results shown in Table 4.

After 35 days of storage, a candle placed inside the silo revealed an average of two (2) live insects per kg of grain. According to equation (16), this equates to a 90% efficiency rate. Since the silo is potentially effective at preventing infestation, it is clear that the new design will

Table 4

Recorded data of fabricated advance silo in grain storage condition (Relative humidity, Ambient temperature, and Average temperature).

Sample (1/4 kg)	Number of live insects	
	Before	After
1st	7	1
2nd	6	1
3rd	8	1
4th	4	0
5th	3	0
6th	6	1
7th	4	0
8th	6	1
9th	6	0
10th	6	1
Average	5.6	0.6

Note: Average live weevils per kg of grain before and after loading = $5.6 \times 4 = 22$ and $0.6 \times 4 = 2$ insects, respectively.

solve the infestation issues with hermetic storage technology (HST) that were highlighted by Baributsa and Njoroge (2020). Thirty (30) live insects per kg were observed on off-loading when the candle was not encased in the silo, which equates to an efficiency of 36% after 35 days of grain storage. This means that after filling the silo with grains, a candle should be placed inside them (Kimani et al., 2018). The reduction in oxygen and increase in the carbon dioxide inside the silo is due to the presence of candle leads the perishing of the present live insects. Because gas exchange between silo and the outside surrounding hinder due to airtightness of the pad locking system. Kimani et al. (2018) found that keeping a candle lit in a storage area kept insects away from the grain, consistent with their findings.

The weekly average temperatures measured by the thermocouple indicated a modest and inconsequential temperature gradient ranging from 25.5 °C to 26.5 °C from the silo wall to the centre and from 24.9 °C to 26.1 °C from the bottom to the top, whereas the ambient temperature varied from 20 °C to 29 °C. Due to significance variation between the interior and outside of the silo and the relatively constant temperature inside, it can be concluded that reinforced natural fibers is an excellent insulator of the silo walls. Adeduntan and Falayi (2017) found that the sawdust-filled double wall greatly decreased temperature variation by 68%. After 15 weeks of storage, the constructed silo showed no signs of grain caking or lumping, making it an efficient grain storage facility. This can be greatly attributed to the good insulation provided by the padlocking and natural fiber reinforcement into the cementitious silo wall. Additionally, it is suggested that the grain's moisture level be examined before to loading to prevent moulds, with results falling within the permitted range of moisture content for grain (12.8%–15.20% wet basis) (Codex Alimentarius Commission, 2017).

3.2. Experimental investigation of safer grain storage parameters

The innovation in the design and construction of small cementitious silos has been demonstrated as a successful solution and efficient technique for decreasing post-harvest grain losses in many countries, especially in developing countries. This technique enhances personal food security by giving small farmers a daily living and financial support. More than two million individuals in Africa are now benefiting from this technology. More than five thousand small advance cementitious silos were successfully implemented as part of a ministry of food and agriculture (MoFA) project in Ghana designed to reduce food waste. The food and agriculture organization (FAO) efforts have also been effective in introducing modern cementitious silos to several other nations (Mobolade et al., 2019).

Among the most significant features of an advanced cementitious silo are its capacity to store between 50 and 1000 kg of grain, its ability to preserve the quality of stored grains, and its elimination of the need for

chemical pesticides. Because the long-term use of which can have a detrimental effect on human health. The advanced cementitious silo is space-efficient and suitable for either residential or agricultural use. As an added benefit, this technology safeguards the farmer from natural climate fluctuations in the price of grains. In this effort, a cementitious silo was modified to be portable and airtight, allowing for the use of effective herbal pesticides and bio-fumigation to protect against rodents and other pests that could be great initiative to overcome post-harvest losses in ecofriendly way. The government of several countries, including India, is now investing in the construction of advance cementitious silos to ensure the long-term security of grain supplies, since grain storage efficiency of silos has grown significantly as a result of technological developments. However, grain storage management and technological advancement are still needed to minimize the post-harvest loss at the small-scale level. Because lack of awareness or proper knowledge about the grain storage management before storing the grain. What the moisture content of grains should be during storage and how it should be calculated are two main factors. As a result, research is required to create management recommendations for safe storage and drying to assure quality management of stored grains. Post-harvest losses will be reduced to a larger extent if collected grains are dried to safe moisture levels (Sharon et al., 2014). Therefore, a test has been performed and store 500 kgs of each of the grains (rice, wheat, maize, and beans) for four weeks each and measure the various parameters at different days as mentioned in Table 3. The grains' moisture content was measured just before being stored in an advanced constructed cementitious silo to ensure that they were within the acceptable range. Wet basis (moisture %) values for rice, wheat, maize, and beans were 9.23%, 11.29%, 13.02%, and 18.50%.

Fig. 9(a) shows that the moisture content of each grain grew from day 1st to 3rd days, then decreased sharply from 3rd to 6th days, and finally stabilized after 6th days. Since the temperature rose from 27 °C to 30 °C between 1st and 3rd days, moisture content was sharply increased. The stored grain's respiration rate increases as the temperature increases, resulting in increased oxygen consumption and carbon dioxide and water release. The water released caused moisture in the grains, increasing moisture levels. The moisture content dropped every day as the temperature dropped from 30 °C until the 12th day when it reached a temperature of 21 °C. This occurred because the respiration of grains slowed to a stream level at low temperatures. The emitted wetness diffused into the surrounding air. After 12th days at 21 °C, the moisture content remained constant because the grains had reached hygroscopic equilibrium.

Since the cementitious wall has a poor thermal conductivity (K) of 0.00502 W/cm°C, the quick decrease in silo temperature with ambient air temperature suggested minimal or very minor change. It assumed that the cementitious wall acted as a thermal conductor, transferring heat from the outside to the inside (or vice versa at a very low rate). This quality helped keep the cementitious wall at a steady (or almost constant) temperature for a considerable time. Additionally, the thermal conductivity (which is just the product of the density and the specific heat) was poor. Due to diffusion or moisture migration (although at a low rate) from the inside to the outer environment, the silo's temperature remained stable. Further, it also observed that the relative humidity was stable from 1st day to 5th day (90%), and then decreased with varying rate (71%–55%) in between 5th day to 16th day As shown in Fig. 9(b). The relative humidity remained stable or near constant at 55% after the 16th day. A shift in the weather caused this. Harmattan like sets in during the experiment was accountable for the consistent temperature.

After four weeks of storage, an inspection revealed no signs of moldiness in the grains. Also, no insect infestation was observed that harmed the grain. Based on the results obtained, the decrease in silo temperature is an indication of the lack of insect activity. This was because insect-infested stocks generated very high temperatures within the silo. Humidity levels below 75% RH are not conducive to mold

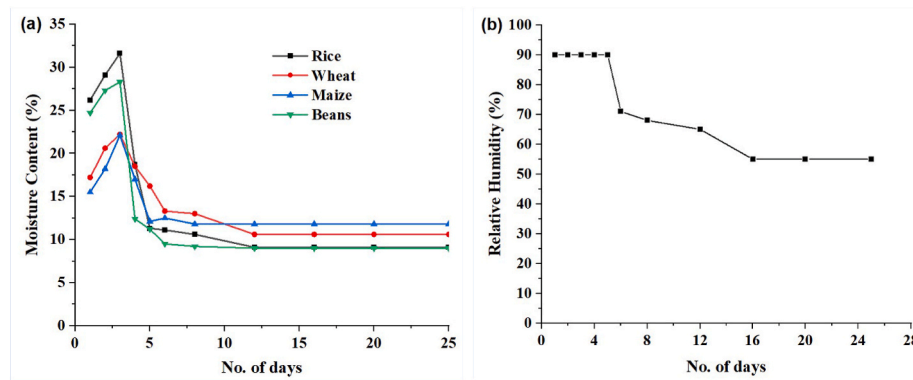


Fig. 9. Variation of (a) Moisture content % and (b) Relative humidity with number of days.

growth or insect activity (Griffiths, 1964). Since the relative humidity in the silo did not stay above 75% for very long, neither mold nor insect growth was detected in the grain.

3.3. Impact of moisture and layer arrangement on silo temperature and pressure distribution

The temperature distribution of wheat grain in the layers near the silo wall is depicted in Fig. 10. The presented data corresponds to the 3rd, 6th, 12th, 18th and 25th day of wheat grain storage. In the initial scenario shown in Fig. 10 (a), with a lower moisture content ($W_L = 9\%$ w. b.) in the lower layer compared to the upper one ($W_U = 18\%$ w. b.), a rapid temperature increase occurred at the silo wall. The upper layer exhibited the highest values due to faster respiration and more intense heat release associated with higher moisture content. In the upper chamber, the temperature rose from 20 to 28 °C after 3rd days, reaching 31 °C after 6th days, 32.5 °C after 12th days, 37.5 °C after 18th days, and 42.5 °C after 25th days. The most significant temperature increase was observed at the border layers, where the difference in moisture contents was the greatest. Conversely, the lower layer experienced a noticeably smaller temperature increase, with slight drops noted at the highest measuring point, attributed to water evaporation from the surface grain layer. From Fig. 10(b), it can be inferred that the temperature increase was less pronounced in reversed layer placement ($W_L = 18\%$ w. b., and $W_U = 9\%$ w. b.). Additionally, higher moisture contents were associated with higher temperature values. Throughout the measurement days, temperatures were consistently lower in the higher sections of the silo. Temperature levels rose with the duration of grain storage. After 25th days, the maximum temperature in the lower part of the silo reached 34 °C, gradually decreasing towards the upper parts to 25.5 °C.

Similar temperature trends were observed for rice (Fig. 11(a)) and beans (Fig. 11(b)). Higher temperatures were documented when the silo

was filled in the first manner (lower layer with lower moisture content than the upper layer). The process of self-heating in rice grain was notably slower than in wheat. The maximum temperature in the upper part of the silo reached 30.5 °C for rice, and the highest temperatures were recorded during beans storage, reaching 49 °C after 25th days. In the case of bean storage, the temperature increased by 29 °C. Rice exhibited the least temperature increase (9 °C). Such substantial temperature changes undoubtedly impact tensions in the silo wall and contribute to the growing difference between grain and ambient temperatures. Therefore, these significant variations should be considered in the silo design process.

Studies demonstrated that, in all scenarios, temperatures at points along the silo axis were 2 °C higher than those at side walls at the same measurement level. Results for material loads are depicted in Figs. 12 and 13. The load on the cylindrical silo wall for all studied grains increased in all measured layers with storage time. The highest load values were observed for wheat, while beans exhibited the lowest. Maximum values typically occurred at the level of 300–500 mm. Fig. 12 (a) illustrates the course of wheat load changes at the lower layer's initial moisture content $W_L = 9\%$ and the upper layer's moisture content $W_U = 16\%$. The highest pressure, measured just after bin filling (1st day), was 1.4 kPa, occurring at 300 mm. It increased to 2.2 kPa after 6th days, 3.2 kPa after 12th days, 3.90 kPa after 18th days, and 4.30 kPa after 25th days of storage. The load increment index reached a value of 3. With the reversed layer arrangement as shown in Fig. 12(b) ($W_L = 18\%$ w. b., $W_U = 9\%$ w. b.), the highest pressure was 1.60 kPa at the level of 400 mm on the 1st day. Still, the most substantial increments occurred in the lower part of the bin (at the level of 100 mm). After 6th, 12th, 18th, and 25th days, the load amounted to: 1.65, 1.75, 2.17, and 2.90, respectively, with a slightly higher index (2.25) observed at 600 mm, despite low pressure values. This was associated with the high hygroscopicity of dried grain. With the first way of layer arrangement ($W_L =$

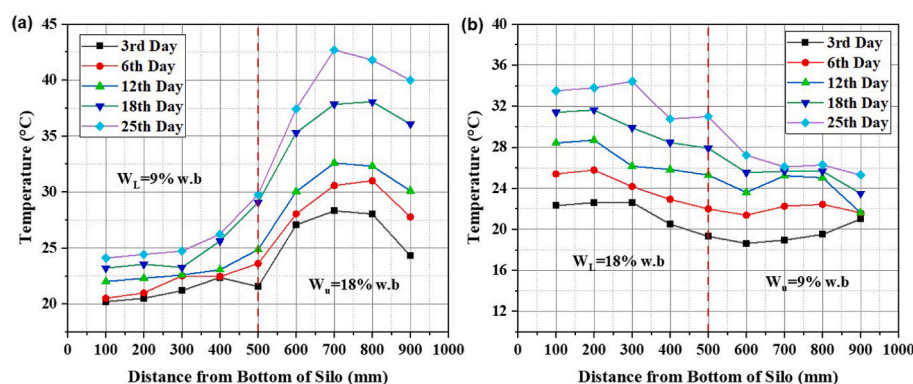


Fig. 10. Wheat temperature variation in silo: moisture impact at (a) $W_U = 18\%$ w. b, $W_L = 9\%$ w. b (b) $W_U = 9\%$ w. b, $W_L = 18\%$ w. b.

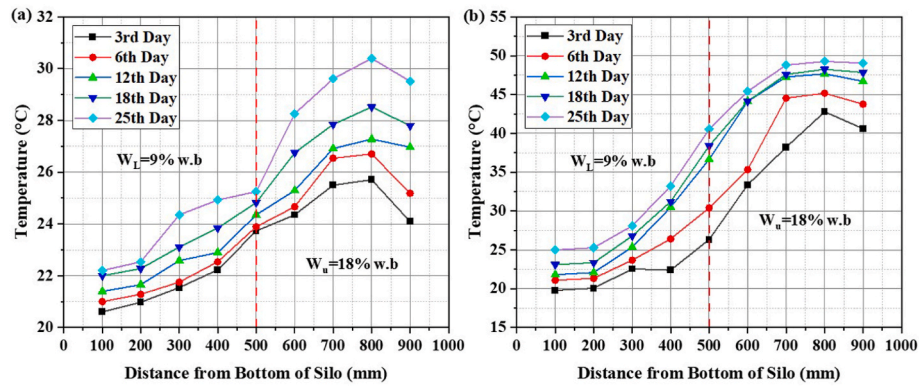


Fig. 11. Temperature at the silo walls depends on the height and storage time at $W_u = 18\%$ w. b and $W_L = 9\%$ w. b for (a) Rice and (b) Beans.

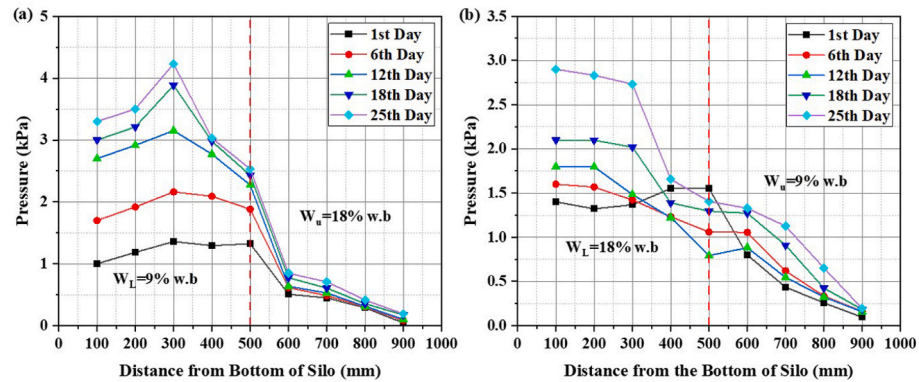


Fig. 12. Silo wall pressure for wheat grain with different moisture content arrangements (a) $W_u = 18\%$ w. b, $W_L = 9\%$ w. b (b) $W_u = 9\%$ w. b, $W_L = 18\%$.

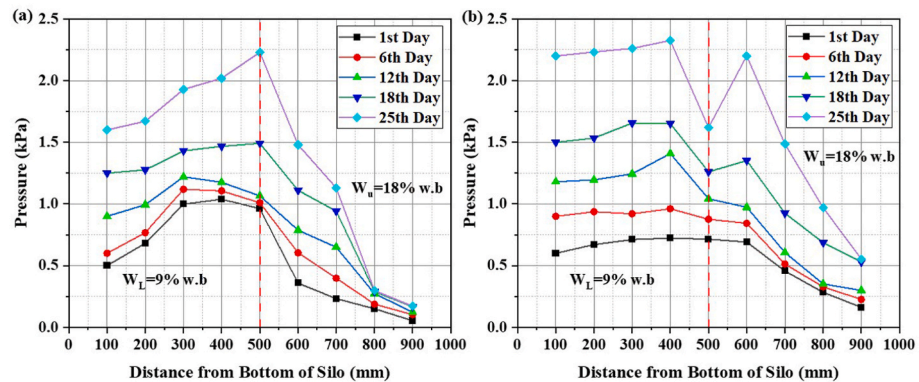


Fig. 13. Silo wall pressure at the different moisture content ($W_u = 18\%$ w. b, $W_L = 9\%$ w. b.) for (a) Rice and (b) Beans.

9% w. b, $W_u = 16\%$ w. b), other grains loads were also higher than with the converse arrangement. They will be further discussed. The values obtained for rice pressure (Fig. 13(a)) were lower than for wheat. The maximum load value on the silo wall for rice occurred at the level of 400 mm on 1st day, amounting to 1.05 kPa. After 6th, 12th, 18th, and 25th days, it increased to 2.25 kPa (increment index 1.95). The highest loads were finally obtained at the level of 500 mm (2.25 kPa). At the level of 500 mm, the pressure increment index reached the value of 4.6, but the load changed only from 1.05 to 2.25 kPa. The process was affected by the diffusion of water absorbed in the upper part of the bin. The lowest load values were obtained for beans (Fig. 13(b)). In this case, the maximum values occurred at the level of 400 mm, increasing from 0.70 to 2.3 kPa, with an increment index of 3.13. Despite low-pressure values, similarly to rice, high indices of pressure increment (3–3.2) were also

obtained in the upper part of the silo at the levels of 600–800 mm. The data shows that when storing rice, wheat, or beans, temperatures increased significantly, leading to higher thermal loads on the bin wall and increased material loads, sometimes by several times. Both parameters could influence wall deformation. The obtained results underwent variance analysis, confirming that temperature and pressure values in the silo were strongly influenced by the type of grains, storage time, height of the measuring point, and arrangement of layers with different moisture contents.

4. Conclusions

The successful design and construction of an indoor silo, attaining 90% effectiveness in controlling insect infestation, showcase promising

results. The research suggests optimal outcomes by enveloping a candle in dry grains, utilizing a cementitious wall filled with natural fibers, and implementing a hermetic padlocking system. These features prevent temperature changes and moisture condensation on the interior wall, curbing caking and lumping of stored grains. However, the relationship between moisture content, temperature, and load on the bin wall during storage emphasizes the significance of grain type and layer arrangement, providing valuable insights for optimizing storage conditions. The portable and improved discharge units also enhance its appeal, while cost analysis reveals affordability, reducing post-harvest losses and increasing farmers' revenue. The silo, ensuring food grain storage throughout the off-season, is poised to enhance food security for small and medium-scale farmers, especially in East Indian states.

The advanced cementitious silo maintains grain quality, enables effective non-residual bio-fumigation, and reduces losses significantly by sealing off rodents and suffocating insects. This enhances stored grains' quantitative and nutritional worth, empowering farmers in market negotiations. It allows farmers to capitalize on seasonal changes in grain prices, ensuring food security during the off-season. Consequently, the advanced cementitious silo positively impacts farmers' wages and living standards. With scalability from 50 kg to 2000 kg, this cutting-edge design holds the potential for adoption in various countries.

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Data availability

No data was used for the research described in the article.

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